

Problem Set 3, Question 2(b)

Long-Run Equilibrium Explained

ECO310: Math Intermediate Microeconomics

Spring 2026

1 The Key Difference: What Happens in the Long Run?

(Lecture 11-12, slides 32-37)

Short-Run Equilibrium (Recap)

Short Run:

- Number of firms N is **FIXED**
- **Only condition:** Supply = Demand
- Firms might make **profits or losses**
- We don't care about entry/exit yet

Long-Run Equilibrium

Long Run:

- Number of firms N **ADJUSTS** (firms can enter/exit)
- **Two conditions:**
 1. Supply = Demand
 2. **All firms make ZERO profit** (break even)
- Price settles at $p = \min AC$ (minimum average cost)

2 Why Zero Profits in the Long Run?

(Lecture 11-12, slides 32-35)

From the slides:

“If firms currently in the market are making money, new firms will come in, shifting supply to the right, thus decreasing the SR equilibrium price”

“If firms are losing money, some will exit, shifting supply to the left, thus increasing the SR equilibrium price”

“These adjustments keep happening until no firms want to enter or exit — only when profits are zero for all firms!”

The Logic:

1. **If $p > \min AC$:** Firms make positive profit $\Rightarrow$ new firms enter $\Rightarrow$ supply increases $\Rightarrow$ price falls
2. **If $p < \min AC$:** Firms make losses $\Rightarrow$ firms exit $\Rightarrow$ supply decreases $\Rightarrow$ price rises
3. **Equilibrium:** $p = \min AC \Rightarrow$ profit = 0 $\Rightarrow$ no incentive to enter or exit

3 Long-Run Equilibrium Conditions

(Lecture 11-12, slide 36)

In long-run equilibrium with identical firms:

LR Equilibrium Conditions:

$$p = MC \quad (\text{profit maximization}) \quad (1)$$

$$p = \min AC \quad (\text{zero profit condition}) \quad (2)$$

This means: $MC = AC$ at the equilibrium quantity!

4 How to Find LR Equilibrium

(Lecture 11-12, slide 37)

Step 1: Find equilibrium quantity per firm

Set $MC = AC$ and solve for q^* :

$$MC(q) = AC(q)$$

Step 2: Find equilibrium price

Evaluate either MC or AC at q^* :

$$p^* = MC(q^*) = AC(q^*)$$

Step 3: Find equilibrium number of firms

Use demand curve with supply = demand:

$$N \cdot q^* = Q^D(p^*)$$

Solve for N :

$$N = \frac{Q^D(p^*)}{q^*}$$

5 Problem 2(b) Solution

Given:

- Market demand: $Q^D(p) = 200 - 5p$
- Each firm has cost: $C(q) = f(N) + \alpha q^2$
- In LR, firms can enter/exit freely

Find:

Each firm's quantity, the price, in LR equilibrium

5.1 Step 1: Find MC and AC

Marginal Cost:

$$MC(q) = \frac{dC}{dq} = 2\alpha q$$

Average Cost:

$$AC(q) = \frac{C(q)}{q} = \frac{f(N) + \alpha q^2}{q} = \frac{f(N)}{q} + \alpha q$$

5.2 Step 2: Find equilibrium quantity per firm

Set $MC = AC$:

$$\begin{aligned} 2\alpha q &= \frac{f(N)}{q} + \alpha q \\ 2\alpha q - \alpha q &= \frac{f(N)}{q} \\ \alpha q &= \frac{f(N)}{q} \\ \alpha q^2 &= f(N) \\ q^* &= \sqrt{\frac{f(N)}{\alpha}} \end{aligned}$$

5.3 Step 3: Find equilibrium price

$$\begin{aligned} p^* &= MC(q^*) = 2\alpha q^* \\ &= 2\alpha \sqrt{\frac{f(N)}{\alpha}} \\ &= 2\sqrt{\alpha f(N)} \end{aligned}$$

Or using AC :

$$\begin{aligned} p^* &= AC(q^*) = \frac{f(N)}{q^*} + \alpha q^* \\ &= \frac{f(N)}{\sqrt{f(N)/\alpha}} + \alpha \sqrt{\frac{f(N)}{\alpha}} \\ &= \sqrt{\alpha f(N)} + \sqrt{\alpha f(N)} \\ &= 2\sqrt{\alpha f(N)} \quad \checkmark \end{aligned}$$

Final Answer for 2(b):

Quantity per firm: $q^* = \sqrt{\frac{f(N)}{\alpha}}$

Price: $p^* = 2\sqrt{\alpha f(N)}$

6 Comparison: SR vs. LR Example

(Lecture 11-12, slides 39-41)

Example from class:

- Each firm: $C(q) = q^2 + 1$
- Demand: $Q^D(p) = 100 - 5p$

Short-Run Equilibrium ($N = 10$ fixed):

- Each firm: $MC = 2q \Rightarrow q = \frac{p}{2}$
- Market supply: $Q^S = 10 \times \frac{p}{2} = 5p$
- Equilibrium: $5p = 100 - 5p \Rightarrow p^{SR} = 10$
- Each firm produces: $q = 5$
- **Number of firms:** $N = 10$ (fixed)

Check profits:

$$\begin{aligned}\pi &= pq - C(q) = 10(5) - (5^2 + 1) \\ &= 50 - 26 = 24 > 0 \quad (\text{making profit!})\end{aligned}$$

Long-Run Equilibrium:

- Set $MC = AC$: $2q = q + \frac{1}{q} \Rightarrow q^* = 1$
- Price: $p^* = MC(1) = 2$
- Market demand: $Q^D(2) = 100 - 5(2) = 90$
- Number of firms: $N = \frac{90}{1} = 90$

Check profits:

$$\begin{aligned}\pi &= pq - C(q) = 2(1) - (1^2 + 1) \\ &= 2 - 2 = 0 \quad (\text{zero profit!})\end{aligned}$$

What happened?

In the SR, 10 firms were making \$24 profit each $\Rightarrow$ new firms entered $\Rightarrow$ price fell from \$10 to \$2 $\Rightarrow$ now 90 firms each make zero profit.

7 Key Takeaways

1. **Short Run:** Fixed $N \Rightarrow$ just set supply = demand $\Rightarrow$ firms might profit/lose
2. **Long Run:** Free entry/exit $\Rightarrow p = \min AC \Rightarrow$ zero profit $\Rightarrow N$ adjusts
3. **LR price is lower:** More firms enter $\Rightarrow$ more competition $\Rightarrow$ lower price
4. **Finding LR equilibrium:** Set $MC = AC$ to find q^* and p^* , then use demand to find N